

I1 extends I2

Interface Inheritance in Java

1. Aim

To demonstrate interface inheritance in Java using "I1 extends I2" and implement the derived interface using the class "C1".

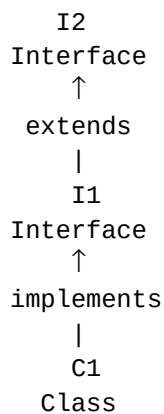
2. Definition

In Java, one interface can inherit another interface using the "extends" keyword.

When "I1 extends I2", interface "I1" inherits all the abstract methods of "I2".

A class implementing "I1" must provide implementations for the methods of both "I1" and "I2".

3. Structure



General Syntax

```
interface I2
{
    void method2();
}

interface I1 extends I2
{
    void method1();
}

class C1 implements I1
{
    public void method1()
    {
        // implementation
    }

    public void method2()
    {
        // implementation
    }
}
```

4. Suitable Example

Here, "I2" represents Person Details, and "I1" extends "I2" to add Student Details.

5. Program

```
interface I2
{
    void displayPerson();
}

interface I1 extends I2
{
    void displayStudent();
}

public class C1 implements I1
{
    public void displayPerson()
    {
        System.out.println("Person details displayed");
    }

    public void displayStudent()
    {
        System.out.println("Student details displayed");
    }

    public static void main(String[] args)
    {
        C1 obj = new C1();

        // Test Case 1
        System.out.println("Test Case 1:");
        obj.displayPerson();

        // Test Case 2
        System.out.println("\nTest Case 2:");
        obj.displayStudent();

        // Test Case 3
        System.out.println("\nTest Case 3:");
        obj.displayPerson();
        obj.displayStudent();
    }
}
```

6. Sample Output

```
Test Case 1:
Person details displayed

Test Case 2:
Student details displayed

Test Case 3:
Person details displayed
Student details displayed
```

7. Test Case Verification

Test Case	Method Called	Expected Output	Result
TC1	displayPerson()	Person details displayed	Passed
TC2	displayStudent()	Student details displayed	Passed
TC3	displayPerson(), displayStudent()	Both details displayed	Passed

Verification

Test Case 1: "displayPerson()" from interface "I2" is executed successfully.

Test Case 2: "displayStudent()" from interface "I1" is executed successfully.

Test Case 3: Both methods are executed successfully, proving that "I1" inherits the method of "I2".

8. Conclusion

Thus, interface inheritance was successfully demonstrated using "I1 extends I2". The class "C1 implements I1" successfully provides implementations for methods declared in both "I1" and its parent interface "I2". All test cases were successfully verified.